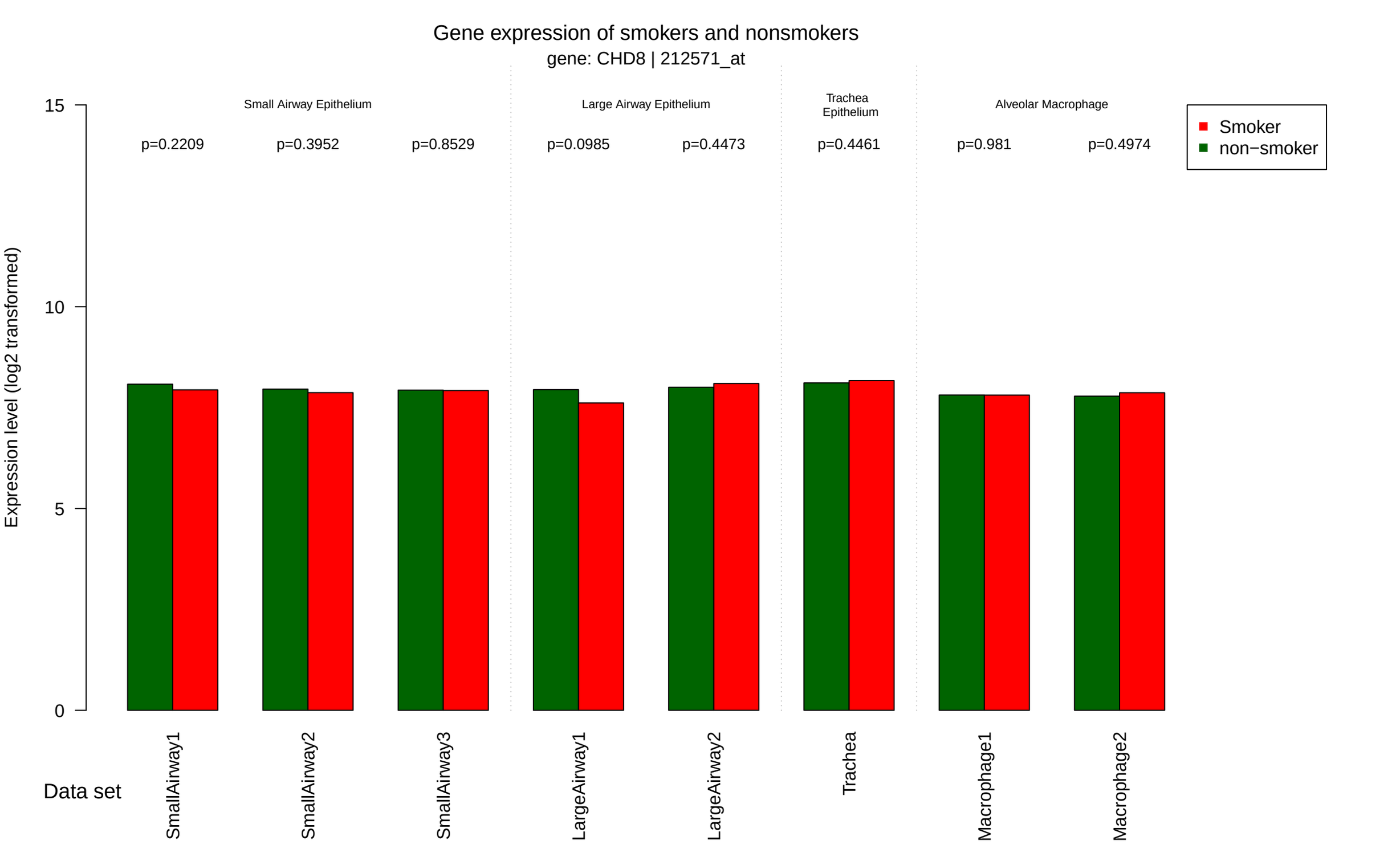
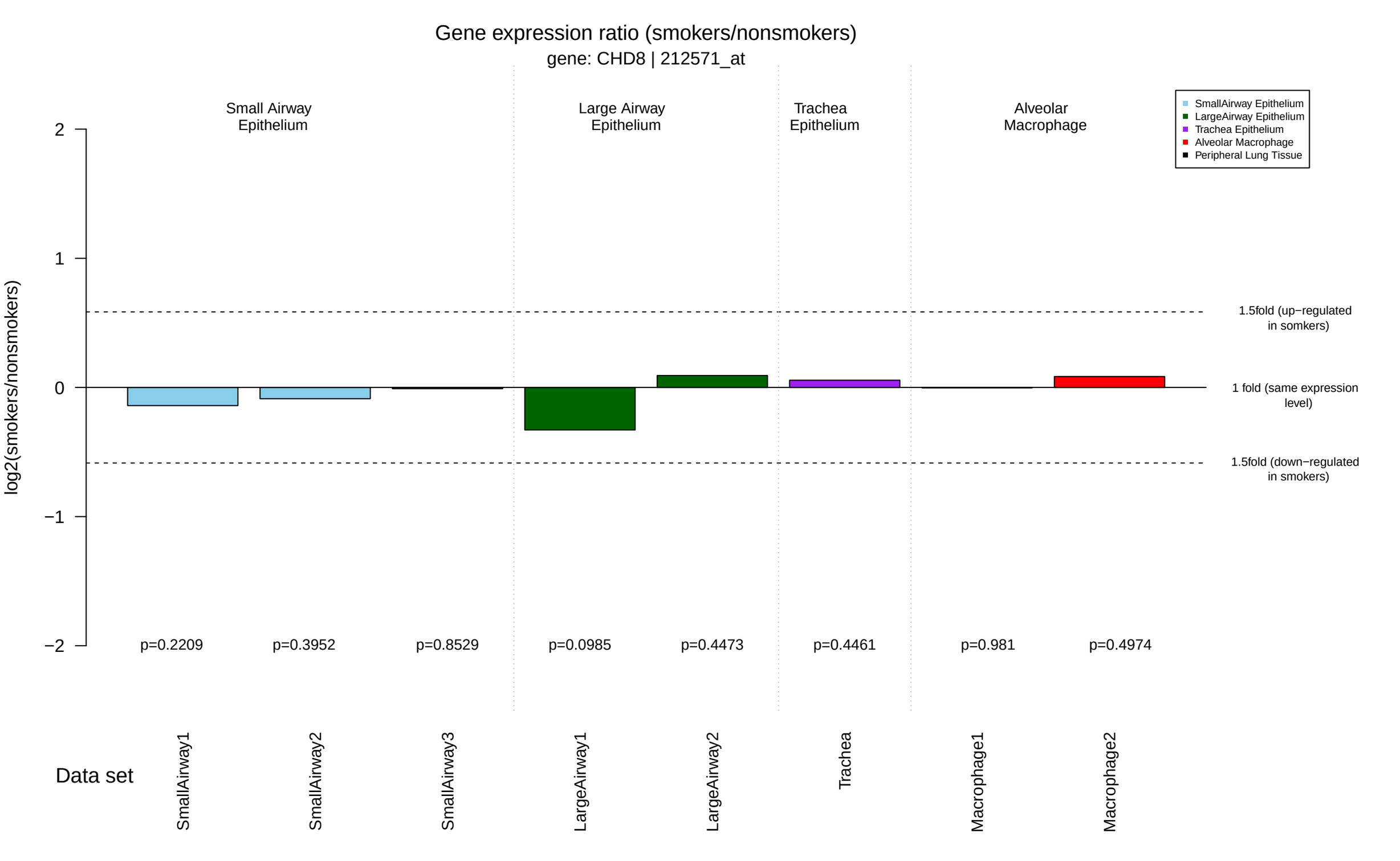
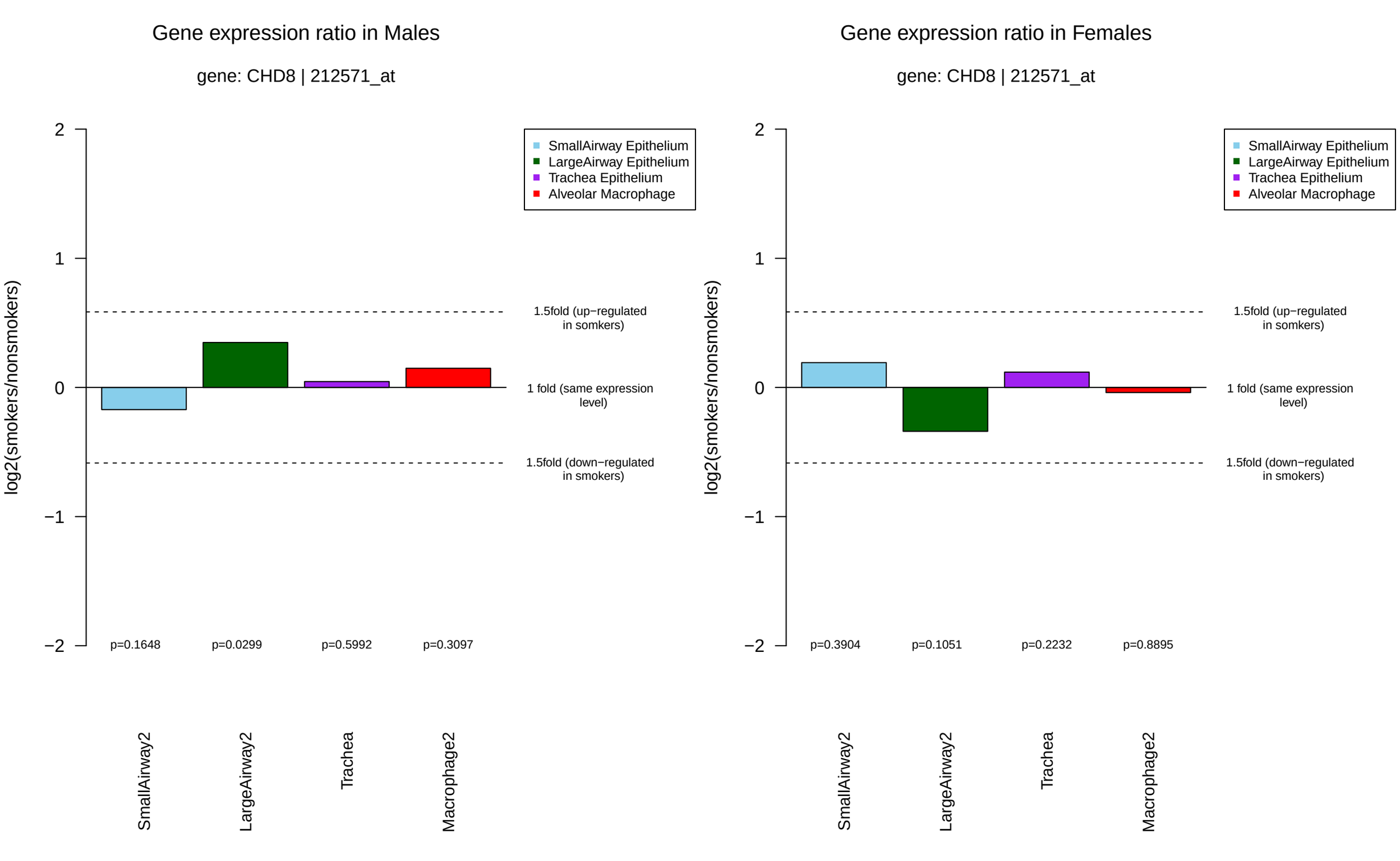
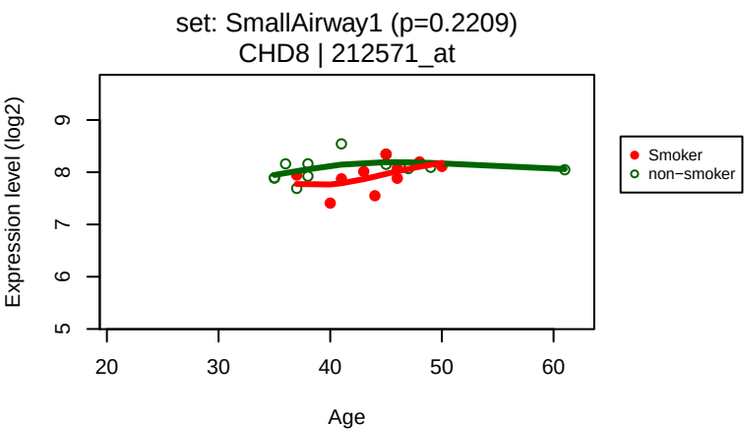
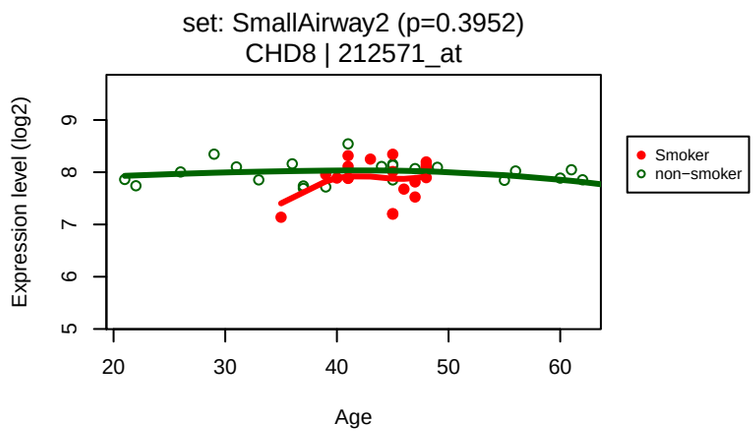
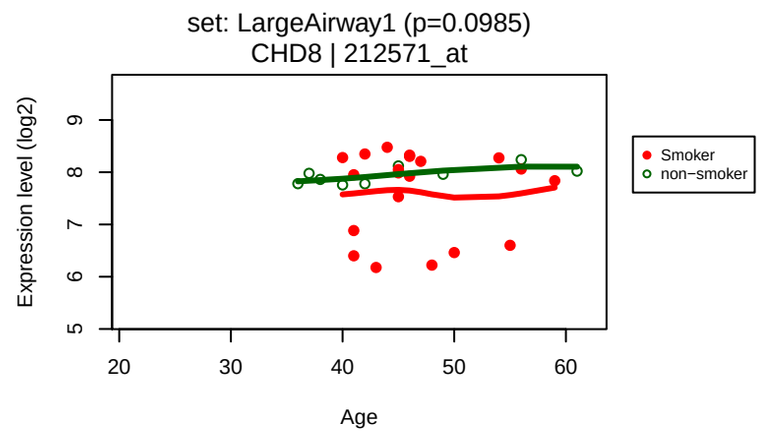

Gene expression of smokers and nonsmokers
gene: CHD8 | 212571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2209
p=0.3952
p=0.8529
p=0.0985
p=0.4473
p=0.4461
p=0.981
p=0.4974
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

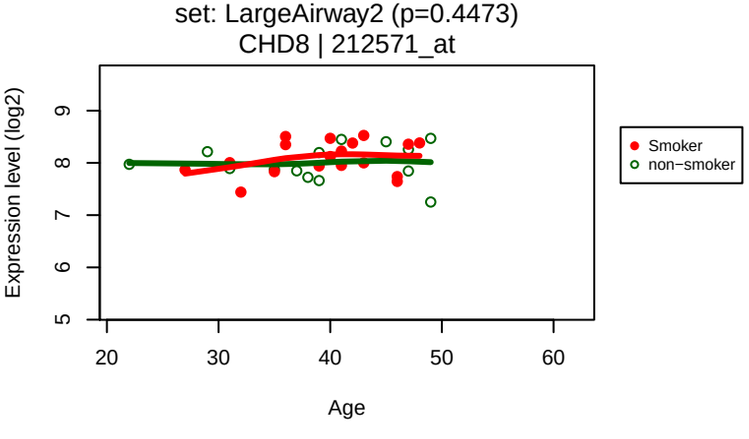

Gene expression ratio (smokers/nonsmokers)
gene: CHD8 | 212571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2209
p=0.3952
p=0.8529
p=0.0985
p=0.4473
p=0.4461
p=0.981
p=0.4974
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

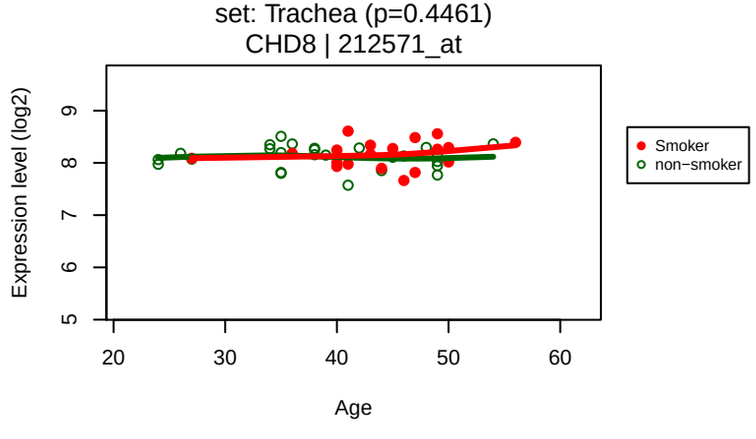

Gene expression ratio in Males
gene: CHD8 | 212571_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1648
p=0.0299
p=0.5992
p=0.3097
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD8 | 212571_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3904
p=0.1051
p=0.2232
p=0.8895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

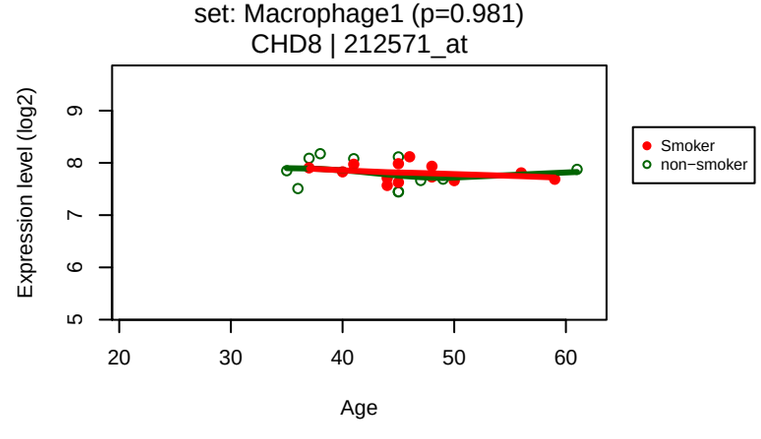

set: SmallAirway1 (p=0.2209)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker

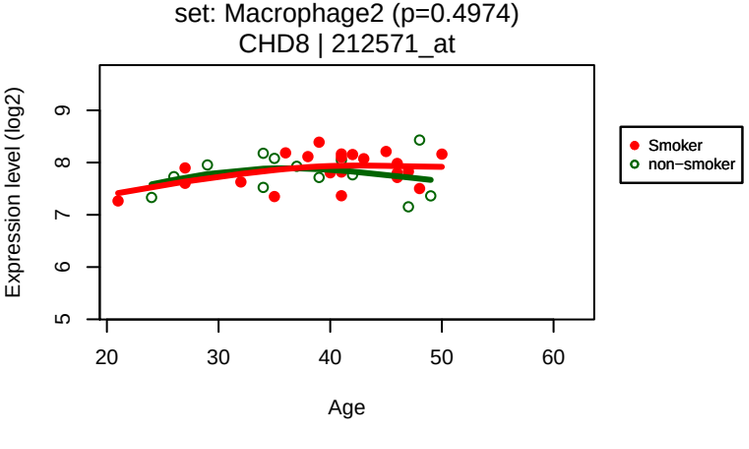

set: SmallAirway2 (p=0.3952)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker

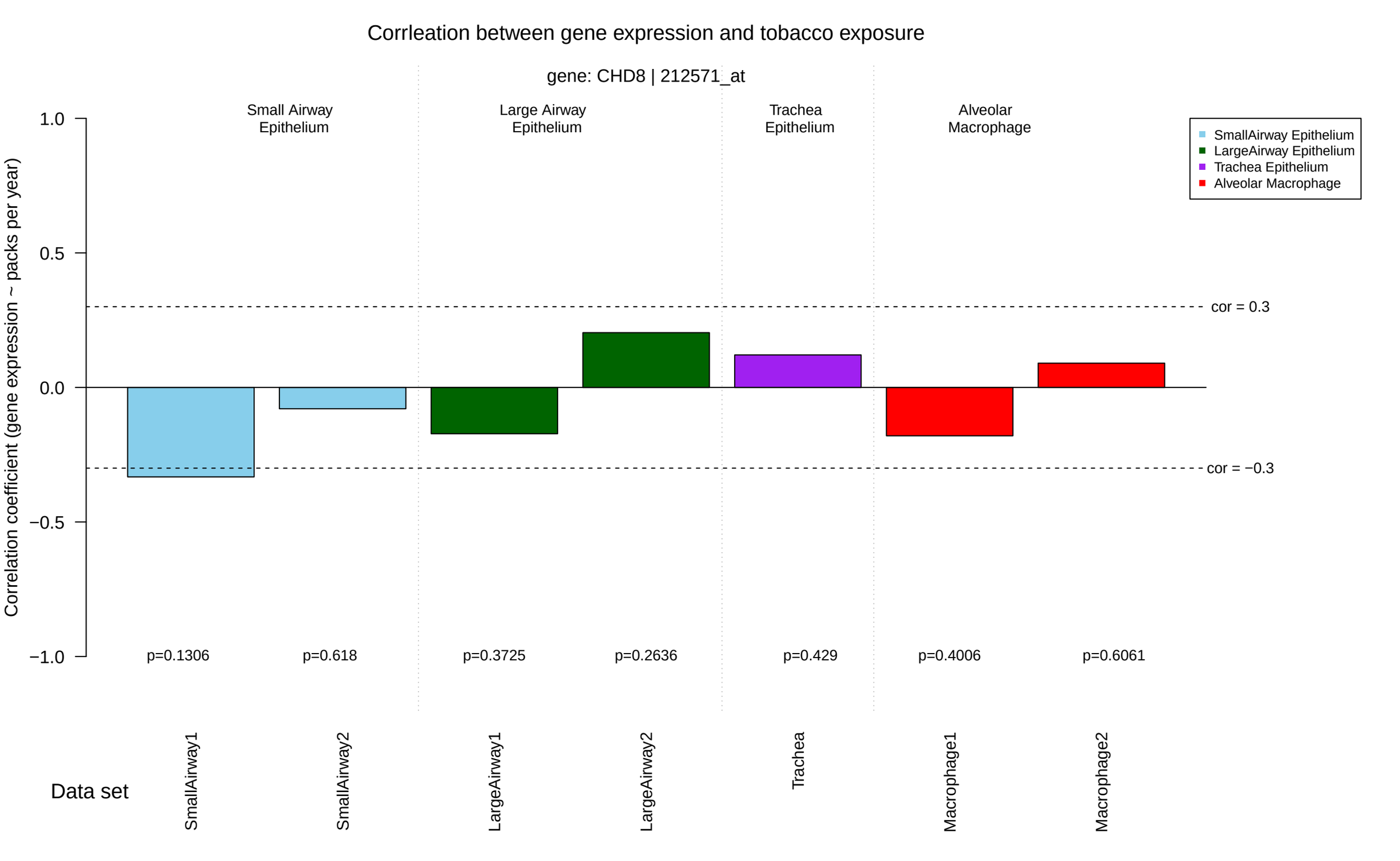

set: LargeAirway1 (p=0.0985)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4473)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4461)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.981)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4974)
CHD8 | 212571_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD8 | 212571_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1306
p=0.618
p=0.3725
p=0.2636
p=0.429
p=0.4006
p=0.6061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2